OFFICIAL AI ENGINEERING ASSESSMENT

NEXYOVI AI WORKSPACE

MASTER HANDBOOK V11.0 — FINAL EDITION

Production-Grade SaaS Engineering Simulation:
Architecture, AI Systems, and Real-Time Infrastructure.

“NOT A TEST — A REAL-WORLD ENGINEERING SIMULATION.”

NEXYOVI TECHNOLOGIES
ENGINEERING DIVISION

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Document Purpose & Scope

The official engineering assessment used to evaluate candidates for the Nexyovi AI Engineering Internship Program.

WHAT THIS ASSESSMENT IS

- **Real-world SaaS Simulation** — an end-to-end engineering challenge designed to simulate production cycles.
- **Full-Stack System Design** — evaluation is based on data integrity, API structure, and frontend execution.
- **AI Integration Eval** — evaluation for neural module reliability and structural data consistency.
- **Production-Level Test** — we evaluate the baseline “Enterprise Readiness” of your code.

WHAT THIS IS NOT

- Not a tutorial or guided programming exercise.
- Not a simple coding task or isolated algorithm test.
- Not a project meant for copy-pasting existing solutions.
- Not a UI-only challenge; backend robustness is prioritized.

EVALUATION PHILOSOPHY

Candidates are evaluated based on system design thinking, logic scalability, AI integration quality, and production readiness. We prioritize maintainable, clean architecture over redundant feature complexity.

KEY EVALUATION AREAS

Architecture Design
Relational schema & API pattern maturity.

AI System Integration
Prompt quality & deterministic data extraction.

Backend Engineering
JWT security & concurrent request handling.

Frontend UX
8px precision & asynchronous loading states.

DISQUALIFICATION NOTICE
Any submission that fails to run locally or lacks core system functionality will be automatically disqualified.

About Nexyovi

Nexyovi Technologies is an AI-first software company building next-generation enterprise systems that combine automation, intelligence, and scalable SaaS infrastructure.

NEXYOVI PAID INTERNSHIP PROGRAM — 12 MONTHS

OFFICIAL 12-MONTH PRODUCTION TRAINING ENVIRONMENT

Successful candidates transition into a high-intensity environment focused on real SaaS ownership.

■ COMPENSATION

- 12-month paid full-time internship.
- Performance-based stipend (reviewed monthly).
- Direct track to Full-Time Employee (FTE).

■ PRODUCTION RESOURCES

- OpenAI Enterprise API access.
- AWS / Cloudinary media infrastructure.
- PostgreSQL / Prisma staging environments.

CAREER PATHWAY

Intern → Junior Engineer → AI Systems Engineer → Full Stack Architect

■ EXPECTATION LEVEL

This is not a tutorial internship. You will be working on real production systems under engineering constraints. Correctness, scalability, and maintainability are prioritized over complexity.

Vision & Overview

MISSION OBJECTIVE

Build the **Nexyovi AI Workspace**: a production-ready multi-tenant SaaS with real-time task management and an AI-powered neural module.

SYSTEM DEFINITION LINE

A multi-tenant SaaS workspace where teams manage tasks, collaborate in real-time, and receive AI-generated productivity insights.

Detailed Specification

3.5.1 CORE MODULES (OBLIGATORY)

1 IDENTITY ENGINE

JWT security with RBAC (Admin/User). Isolated database multi-tenancy.

2 KANBAN ENGINE

Persistent drag-and-drop state management with Status/Priority tiers.

3 AI NEURAL MODULE

AI analysis for priority scores. MANDATE: AI output must be deterministic in structure but dynamic in content.

4 REAL-TIME SYNC

Socket.io event broadcasting. NO POLLING ALLOWED.

STRUCTURED AI CONTRACT

```
{
  "priority_score": 1-100,
  "summary": "AI logic string",
  "suggested_due_date": "ISO format",
  "reasoning": "Decision logic"
}
```

Resilience: AI failure must never affect system availability or user experience. Fallback to rule-based prioritization gracefully.

System Architecture

FRONTEND
Next.js 14 (App Router), TypeScript, Tailwind, Zustand.

BACKEND
Node.js, Service-Repo Pattern, JWT Auth.

■ SYSTEM BEHAVIOR RULE
All system modules must be production-safe, modular, and independently testable.

NON-FUNCTIONAL REQUIREMENTS

- **Support multiple users simultaneously** with concurrency safety.
- **UI must remain fluid and responsive** under load.
- **Database queries optimized for scale** (no N+1 query patterns).
- **Atomic real-time sync** with zero event duplication.

Submission Protocol

Only submissions that are fully runnable locally and deployed will be evaluated.

SUBMISSION REQUIREMENTS
- **GitHub Repo:** separated /frontend and /backend folders.
- **README.md:** architecture docs, AI logic explanation, and setup guide.
- **Live URLs:** required public instances (Vercel/Railway).

 # SETUP EXPECTATION
Backend & Frontend initialization must complete in < 2 minutes.

Evaluation Rubric

CATEGORY	WEIGHT	EXCELLENT BENCHMARK
Code Quality	**30%**	DRY, SOLID, type-safe architecture.
Systems Design	**20%**	Relational integrity & separation of concerns.
AI Integration	**20%**	Structured extraction & prompt engineering.
UI Execution	**20%**	Polished, responsive design on 8px grid.
Documentation	**10%**	Clear setup, defense, and architecture manual.

“Build systems that think, scale, and execute.”

NEXYOVI TECHNOLOGIES
ENGINEERING DIVISION | 2026 INTAKE

This assessment simulates real-world SaaS engineering conditions. Candidates should treat this as a production system, not a tutorial exercise.

ENGINEERING LEADERSHIP BOARD — FINAL RELEASE